

Automated Dynamic Analysis Signature Generation

Using LLMs for Malware Behavior Understanding and CAPEv2 Signature Creation

Ikram Benfellaoui Fadel Fatima Zahra

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Presentation Roadmap

- ① Problem and objective
- ② Our 6-stage pipeline
- ③ Key engineering decisions
- ④ Results and evaluation
- ⑤ Practical impact, limitations, and conclusion

Why This Project Matters

Manual workflow today

- Signature writing is slow
- Inconsistent quality between analysts
- Hard to scale to large malware volumes
- Repetitive effort for similar families

Target workflow

- AI-assisted signature drafting
- Faster analyst turnaround
- Standardized detection logic
- Human validation before deployment

Core question: Can LLMs generate useful CAPEv2 signatures with strong precision and practical analyst value?

Project Scope

- Dataset: 100 malware samples across 10 families
- Input: CAPEv2 dynamic analysis JSON reports
- Output: Python CAPEv2 signatures + MITRE ATT&CK mappings
- Constraint: Low-cost approach using local Llama inference

Deliverables

6 notebooks, processed outputs, 20 generated signatures, evaluation report, and containerized setup.

6-Stage System Pipeline

- ① Data acquisition
- ② Feature extraction
- ③ Behavior IR construction
- ④ LLM behavior analysis
- ⑤ Signature generation and validation
- ⑥ Final evaluation and reporting

Design principle: separate data normalization from LLM reasoning to improve reliability.

Extracted behavior groups

- Process and injection activity
- Registry persistence indicators
- File operations and drops
- Network and command behavior

Behavior IR value

- Normalizes inconsistent report structures
- Reduces prompt noise for the LLM
- Improves reproducibility across runs

LLM and Signature Generation Strategy

- Llama used for semantic interpretation (local, low cost)
- Family-level aggregation to detect recurring patterns
- CAPEv2 signature generation with strict validation:
 - syntax checks
 - logic checks
 - manual analyst review

Important result: Validation caught approximately 15% of generated signature issues before final output.

Main Challenges and Fixes

Challenge	How we solved it
Inconsistent JSON structures	Built robust parsing + Behavior IR abstraction
LLM output quality drift	Added cache, tighter prompts, post-checks
Signature correctness gaps	Added syntax + logic + manual validation
Lower ATT&CK edge-case accuracy	Added confidence-aware review workflow

What Did Not Work Initially

- Feeding raw CAPEv2 JSON directly to Llama
- Fully automated ATT&CK mapping without confidence checks
- Treating generated Python as deployment-ready without validation

Key lesson

Automation in security must include explicit validation gates. Speed without verification creates detection risk.

Primary Performance Metrics

Signature Quality

- Precision: 86%
- Recall: 79%
- F1-score: 0.82
- False positive rate: 8%

Output Volume

- 20 CAPEv2 signatures
- 10 malware families covered
- 35+ ATT&CK techniques mapped

Generalization and Mapping Quality

- Cross-family precision on unseen families: 77%
- Cross-family recall on unseen families: 65%
- ATT&CK mapping overall accuracy: 84%

Interpretation

The model captures reusable behavior patterns, but still requires analyst oversight for edge cases such as subtle Discovery and Lateral Movement indicators.

- Practical analyst acceleration: from hours to minutes for initial signature drafts
- Repeatable pipeline with containerized setup
- Cost-aware execution using local inference
- Strong baseline for SOC workflow augmentation

Bottom line: this is an analyst multiplier, not a full analyst replacement.

Final Takeaways

- ① LLM-assisted CAPEv2 signature generation is feasible and useful.
- ② A Behavior IR layer is critical for stability and quality.
- ③ Validation is mandatory for safe security automation.
- ④ The system provides immediate practical value with clear known limits.

Future Improvements

- Expand dataset size and family diversity
- Improve low-confidence ATT&CK mapping with hybrid rules
- Add automated regression tests for signatures
- Integrate with SOC tooling for continuous feedback

Thank you

Questions?

Repository: <https://github.com/ykram051/Automated-Dynamic-Analysis-Signature-Generation>